

March 11, 2026

recall:

Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable at $\vec{a} \in X$, then

1) The Hessian of f at $\vec{a}$ is $Hf|_{\vec{a}} = D(\vec{\nabla} f)|_{\vec{a}}$ (the matrix of all 2^{nd} order partial derivatives of f at $\vec{a}$.)

2) The 2^{nd} order Taylor polynomial of f at $\vec{a}$ is $p_2(\vec{x}) = f(\vec{a}) + \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a}) + \frac{1}{2}(\vec{x} - \vec{a})^T Hf|_{\vec{a}} (\vec{x} - \vec{a})$

matrix product

"row vector" (1×n) n×n matrix "column vector" (n×1)

e.g. (from end of last class) $f(x, y) = 4x^3y + 3xy^2$ find p_2 at $\vec{a} = (1, 2)$

$$\vec{\nabla} f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (12x^2y + 3y^2, 4x^3 + 6xy)$$

$$Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 24xy & 12x^2 + 6y \\ 12x^2 + 6y & 6x \end{bmatrix}$$

$$f(\vec{a}) = 4(1^3)(2) + 3(1)(2^2) = 20 \quad Hf|_{\vec{a}} = \begin{bmatrix} 24(1)(2) & 12(1^2) + 6(2) \\ 12(1^2) + 6(2) & 6(1) \end{bmatrix} = \begin{bmatrix} 48 & 24 \\ 24 & 6 \end{bmatrix}$$

$$\vec{\nabla} f|_{\vec{a}} = (12(1^2)(2) + 3(2^2), 4(1^3) + 6(1)(2)) = (36, 16)$$

$$\begin{aligned} p_2(\vec{x}) &= 20 + (36, 16) \cdot (x-1, y-2) + \frac{1}{2}(x-1, y-2) \begin{bmatrix} 48 & 24 \\ 24 & 6 \end{bmatrix} \begin{pmatrix} x-1 \\ y-2 \end{pmatrix} \\ &= 20 + 36(x-1) + 16(y-2) + \frac{1}{2}(x-1, y-2) \begin{pmatrix} 48(x-1) + 24(y-2) \\ 24(x-1) + 6(y-2) \end{pmatrix} \end{aligned}$$

$$p_2(\vec{x}) = 20 + 36(x-1) + 16(y-2) + \frac{1}{2} [48(x-1)^2 + 24(x-1)(y-2) + 24(x-1)(y-2) + 6(y-2)^2]$$

e. g. : $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ twice differentiable at $\vec{a} = (1, 2, -1)$
 compute $p_2(\vec{x})$ given that

$$f(1, 2, -1) = 8 \quad \vec{\nabla} f|_{(1, 2, -1)} = (3, 0, 5) \quad \text{and} \quad Hf|_{(1, 2, -1)} = \begin{bmatrix} 4 & 7 & 0 \\ 7 & 5 & -2 \\ 0 & -2 & 0 \end{bmatrix}$$

Write p_2 as a polynomial (you can keep $(x-1)$
 $(y-2)$, $(z+1)$ factors)

$$\begin{aligned} p_2(\vec{x}) &= f(\vec{a}) + \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a}) + \frac{1}{2} (\vec{x} - \vec{a})^T Hf|_{\vec{a}} (\vec{x} - \vec{a}) \\ &= 8 + (3, 0, 5) \cdot (x-1, y-2, z+1) \\ &\quad + \frac{1}{2} (x-1, y-2, z+1) \begin{bmatrix} 4 & 7 & 0 \\ 7 & 5 & -2 \\ 0 & -2 & 0 \end{bmatrix} \begin{pmatrix} x-1 \\ y-2 \\ z+1 \end{pmatrix} \end{aligned}$$

$$= 8 + 3(x-1) + 0 + 5(z+1)$$

$$\begin{aligned} &+ \frac{1}{2} [4(x-1)^2 + (2)7(x-1)(y-2) + 2(0)(x-1)(z+1) \\ &\quad + 5(y-2)^2 + (2)(-2)(y-2)(z+1) + 0(z+1)^2] \end{aligned}$$

$$\begin{aligned} p_2 &= 8 + 3(x-1) + 5(z+1) + 2(x-1)^2 + 7(x-1)(y-2) + \frac{5}{2}(y-2)^2 \\ &\quad - 2(y-2)(z+1) \end{aligned}$$

Extremal points

def: Let $f: \overset{\text{open}}{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $\vec{a} \in X$, we say that

1) $f(\vec{a})$ is a local minimum of f (or that f has a local minimum at $\vec{a}$)

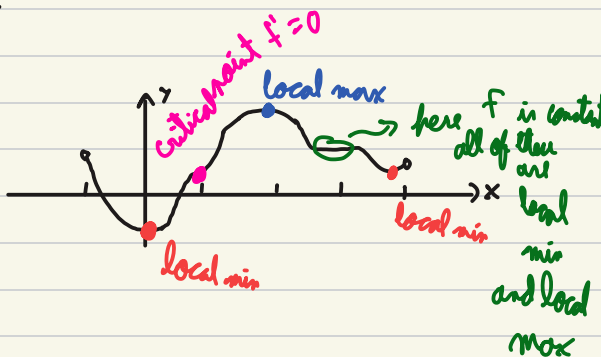
if there is $r > 0$ such that $f(\vec{a}) \leq f(\vec{x})$
for all $\vec{x} \in X$ satisfying $\|\vec{x} - \vec{a}\| < r$.

2) $f(\vec{a})$ is a local maximum of f (or that f has a local maximum at $\vec{a}$)

if there is $r > 0$ such that $f(\vec{a}) \geq f(\vec{x})$
for all $\vec{x} \in X$ satisfying $\|\vec{x} - \vec{a}\| < r$.

3) A local max or local min is called a local extremum
(in plural, local extrema).

e.g. $f(x): (-1, 4) \rightarrow \mathbb{R}$



def (continued)

4) f has a global (or absolute) min. at $\vec{a}$ if $f(\vec{a}) \leq f(\vec{x})$ for all $\vec{x} \in X$.

5) f has a global (or absolute) max. at $\vec{a}$ if $f(\vec{a}) \geq f(\vec{x})$ for all $\vec{x} \in X$.

Theorem (First derivative test): Suppose $f: \overset{\text{open}}{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ has a local extremum at $\vec{a}$. Then,
if f is differentiable at $\vec{a}$ we must have $\vec{\nabla} f|_{\vec{a}} = \vec{0}$.

def: Let $f: \overset{\text{open}}{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$.

A point $\vec{a} \in X$ is called a critical point of f if either:
1) f is not differentiable at $\vec{a}$, or
2) $\vec{\nabla} f|_{\vec{a}} = \vec{0}$

* Critical points are the only points where f could/might have a local extremum.

e.g. Critical points of $f(x, y) = y/x$ $X = \{(x, y) : x \neq 0\}$

$$\vec{\nabla} f = \left(-\frac{y}{x^2}, \frac{1}{x} \right)$$

* $x = 0$ is not in the domain, so it is not a critical point

f is differentiable on all X

$\vec{\nabla} f = \vec{0}$? would need $\frac{1}{x} = 0$ impossible

$\Rightarrow$ f has no critical points

so it can't have any local extrema

e.g. Critical point of $f(x,y) = 1 - 2x^2 - 3y^4 - 4y^3$

$$\vec{\nabla} f = (-4x, -12y^3 - 12y^2) \quad f \text{ is diff on all } \mathbb{R}^2$$

$$\vec{\nabla} f = 0? \Rightarrow -4x = 0 \text{ so } x = 0$$

$$\text{and } -12y^2(y+1) = 0 \text{ so } y = 0 \text{ or } y = -1$$

critical points: $(0, 0)$ and $(0, -1)$



We can use the Hessian to determine the type of critical point

def: Let H be an $n \times n$ matrix:

1) we say H is positive definite if

$$\vec{x}^T H \vec{x} = \vec{x} \cdot (H \vec{x}) > 0 \quad \text{for all } \vec{x} \in \mathbb{R}^n \setminus \{\vec{0}\}$$

2) we say H is negative definite if

$$\vec{x}^T H \vec{x} = \vec{x} \cdot (H \vec{x}) < 0 \quad \text{for all } \vec{x} \in \mathbb{R}^n \setminus \{0\}$$

e.g. Take $H = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ is positive def. for any $\vec{x} = (x, y)$

$$\vec{x} \cdot (H \vec{x}) = (x, y) \cdot (4x, 2y) = 4x^2 + 2y^2$$

if either x or y are not 0 (i.e. $\vec{x} \neq \vec{0}$) $\vec{x} \cdot (H \vec{x}) > 0$

e.g. $H = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix}$ this is negative definite

e.g. $H = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ this is neither pos. def nor neg. def.

Theorem (second derivative test): Let $f: \overset{\text{open}}{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable at $\vec{a}$. Suppose that $\vec{\nabla} f|_{\vec{a}} = \vec{0}$, then

- 1) If $Hf|_{\vec{a}}$ is positive definite, then $f(\vec{a})$ is a local min.
- 2) If $Hf|_{\vec{a}}$ is negative definite, then $f(\vec{a})$ is a local max.

e.g. back to $f(x, y) = 7 - 2x^2 - 3y^4 - 4y^3$

$$\vec{\nabla} f = (-4x, -12y^3 - 12y^2) \quad \text{critical points } (0,0), (0,-1)$$

$$Hf = D(\vec{\nabla} f) = \begin{bmatrix} -4 & 0 \\ 0 & -36y^2 - 24y \end{bmatrix}$$

consider $(0,0)$: $Hf|_{(0,0)} = \begin{bmatrix} -4 & 0 \\ 0 & 0 \end{bmatrix}$ we can't tell from 2nd der. test if this is local extrema
this is neither pos. def. nor neg. def.

$$\text{at } (0,-1): Hf|_{(0,-1)} = \begin{bmatrix} -4 & 0 \\ 0 & -36+24 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -12 \end{bmatrix}$$

$Hf|_{(0,-1)}$ is negative definite

$\Rightarrow f(0,-1)$ is a local max.